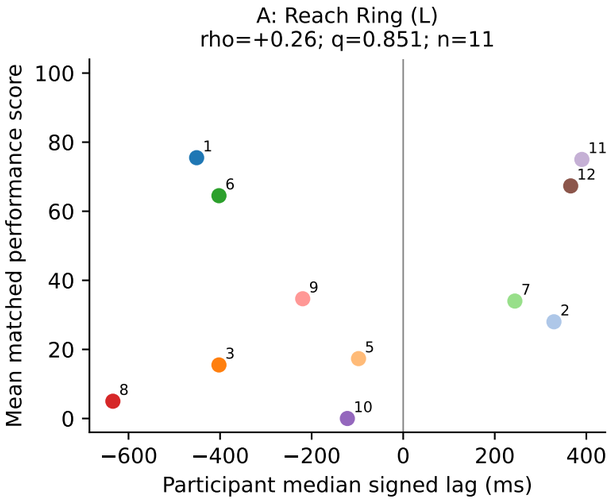
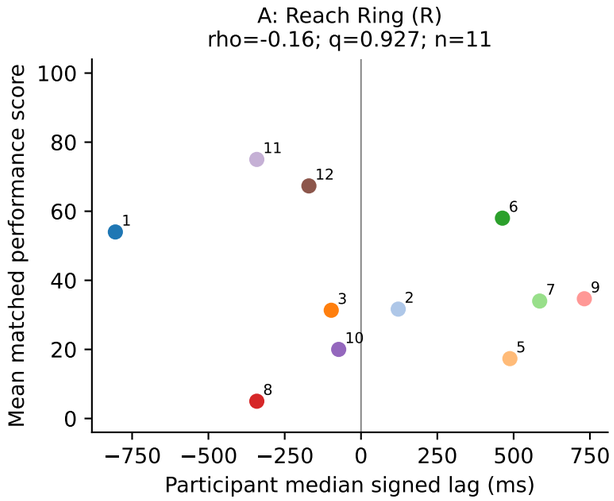
Performance score vs gaze/PSM peak features (position)
Positive signed lag = gaze leads; * BH q < 0.05 within each analysis panel

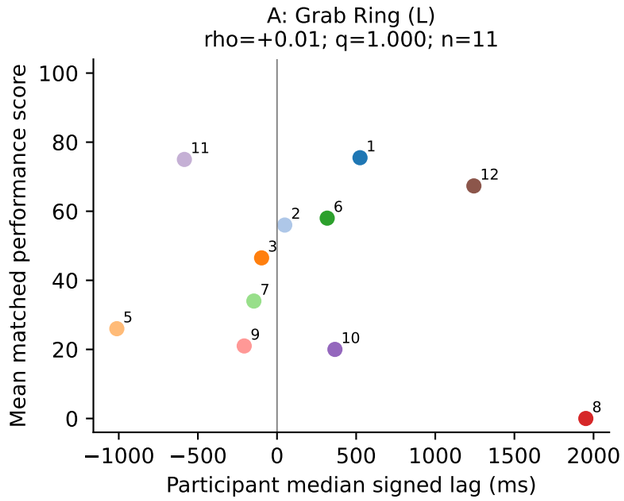
	Between participants			Within participant			Within participant adjusted for task order + attempt		
	Signed lag	Absolute lag	Peak r	Signed lag	Absolute lag	Peak r	Signed lag	Absolute lag	Peak r
A: Whole task (L)	-0.05	+0.03	+0.66	+0.43	+0.55	-0.02	+0.51	+0.48	-0.07
A: Whole task (R)	+0.30	+0.30	-0.43	+0.35	+0.35	+0.12	-0.06	-0.06	+0.15
A: Reach Ring (L)	+0.26	+0.13	+0.43	-0.16	-0.24	+0.26	-0.11	-0.22	+0.32
A: Reach Ring (R)	-0.16	+0.24	+0.04	+0.12	+0.33	+0.15	+0.35	+0.11	-0.03
A: Grab Ring (L)	+0.01	-0.15	+0.27	-0.21	+0.03	+0.13	-0.48	+0.10	+0.18
A: Grab Ring (R)	-0.27	-0.16	+0.52	+0.43	+0.20	+0.17	+0.02	+0.19	+0.36
A: Transfer Ring (L)	+0.00	-0.08	+0.57	+0.35	+0.54	+0.34	+0.35	+0.40	+0.46
A: Transfer Ring (R)	-0.78	-0.79	+0.21	+0.47	+0.38	-0.02	+0.16	+0.07	-0.01
A: Place Ring (L)	+0.01	-0.32	+0.01	-0.03	+0.02	+0.32	+0.15	-0.19	+0.25
A: Place Ring (R)	+0.18	+0.25	+0.08	+0.03	+0.44	-0.36	-0.21	+0.32	-0.33
A: Drop Ring (L)	excluded	excluded	excluded	excluded	excluded	excluded	excluded	excluded	excluded
A: Drop Ring (R)	excluded	excluded	excluded	excluded	excluded	excluded	excluded	excluded	excluded
C: Whole task (L)	-0.24	+0.29	-0.30	+0.05	-0.16	+0.26	-0.03	-0.20	+0.17
C: Whole task (R)	+0.06	-0.44	+0.62	-0.23	+0.15	-0.00	-0.23	+0.23	-0.13
C: Reach String (L)	-0.08	+0.02	-0.50	+0.21	-0.11	-0.18	+0.13	-0.08	-0.10
C: Reach String (R)	-0.12	+0.00	+0.22	-0.14	-0.19	-0.08	-0.14	-0.23	-0.03
C: Threading (L)	-0.20	+0.61	-0.41	low N	low N	low N	low N	low N	low N
C: Threading (R)	+0.08	-0.41	-0.58	low N	low N	low N	low N	low N	low N
C: Wrap around PSM (L)	-0.40	-0.73	+0.44	+0.06	-0.03	-0.01	-0.08	-0.04	-0.25
C: Wrap around PSM (R)	+0.30	-0.56	+0.56	+0.17	+0.16	+0.17	+0.06	-0.13	-0.08
C: Pull/cinch (L)	-0.36	+0.71	+0.29	low N	low N	low N	low N	low N	low N
C: Pull/cinch (R)	excluded	excluded	excluded	excluded	excluded	excluded	excluded	excluded	excluded

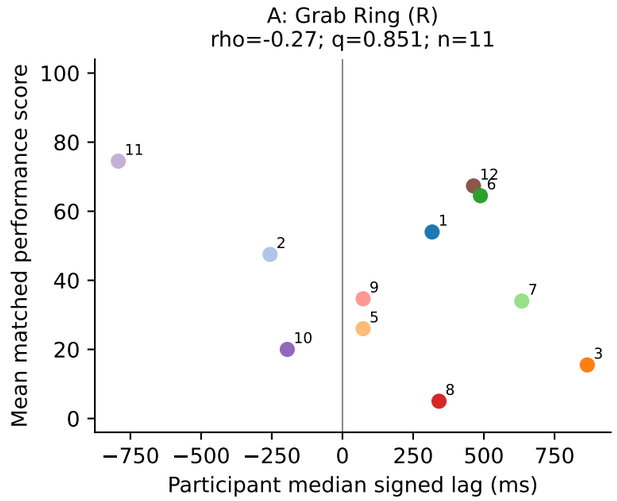
Rank association
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

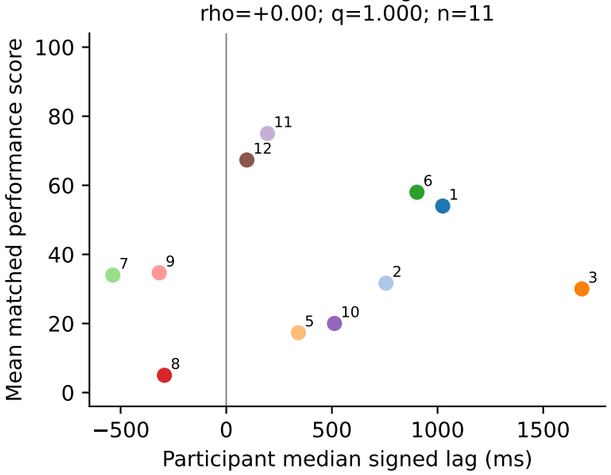
Valid phases: timing vs performance between participants
One point per participant; labels are participant IDs. Task A and C remain separate.

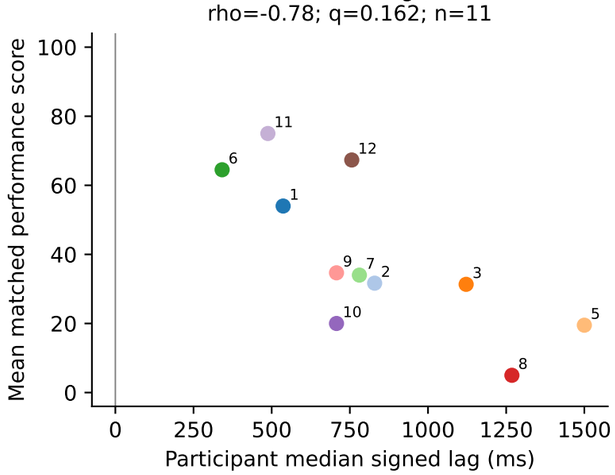


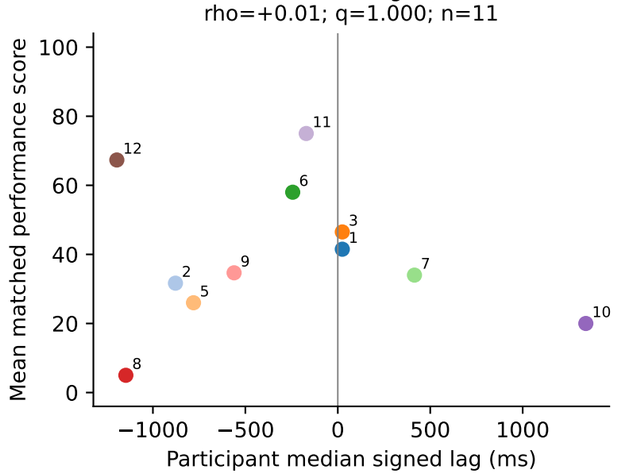


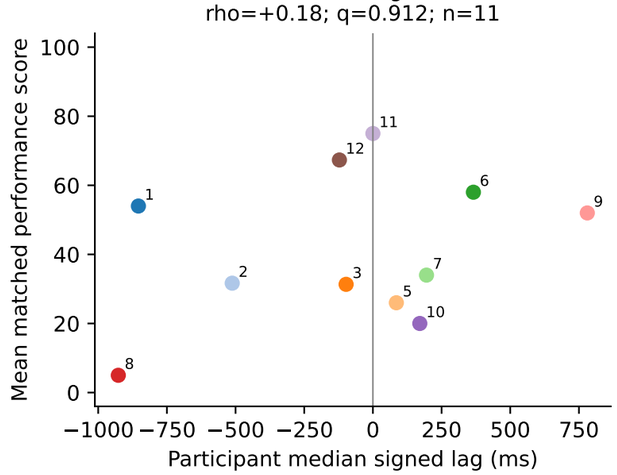


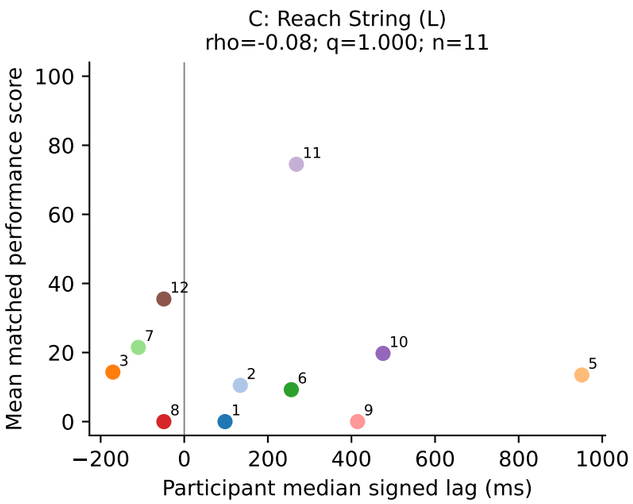


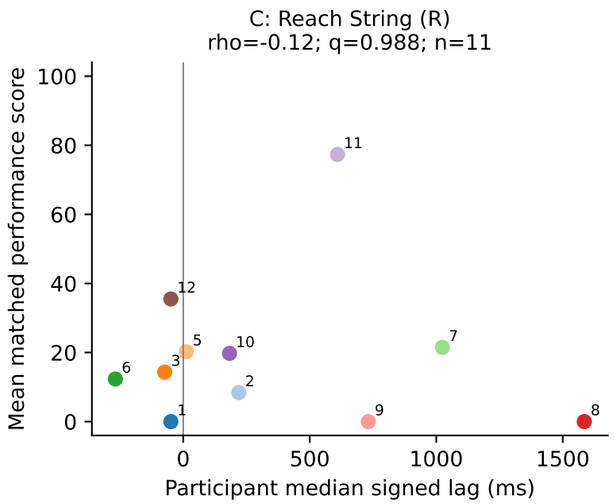


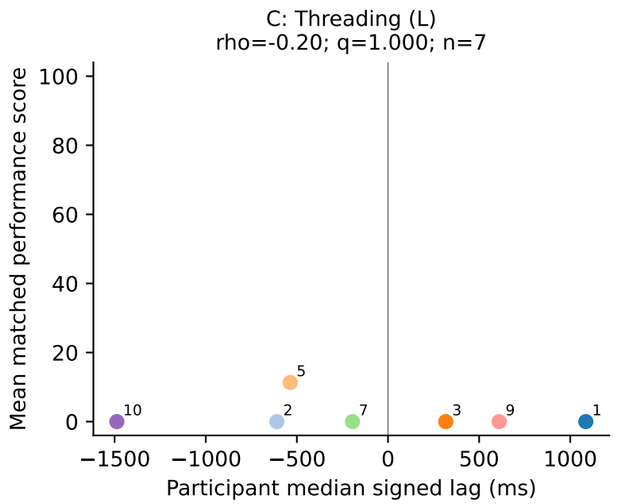


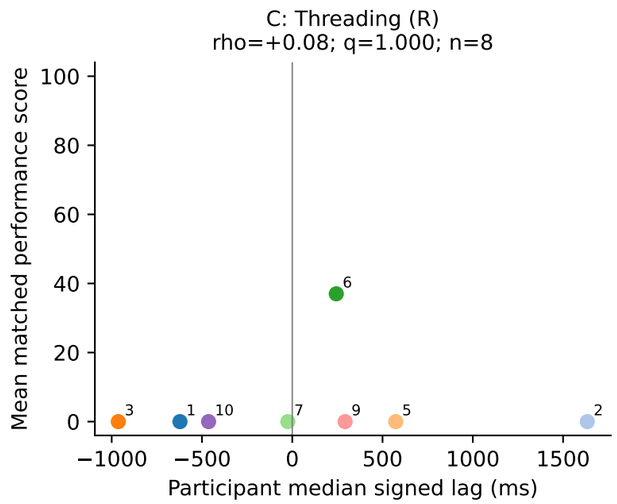


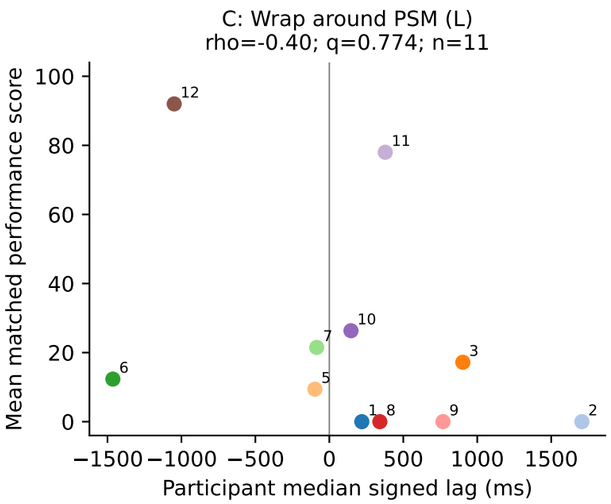


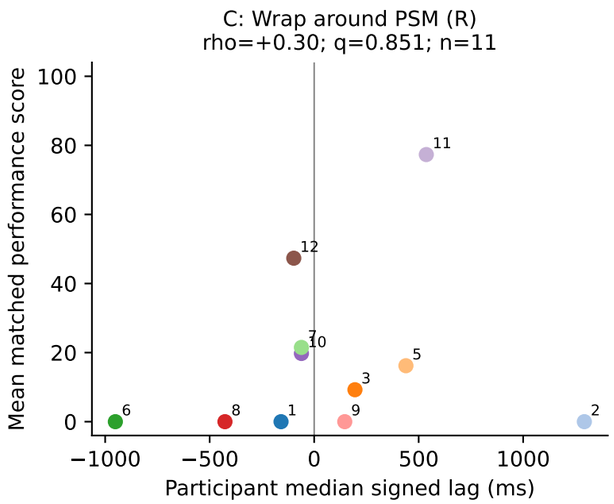


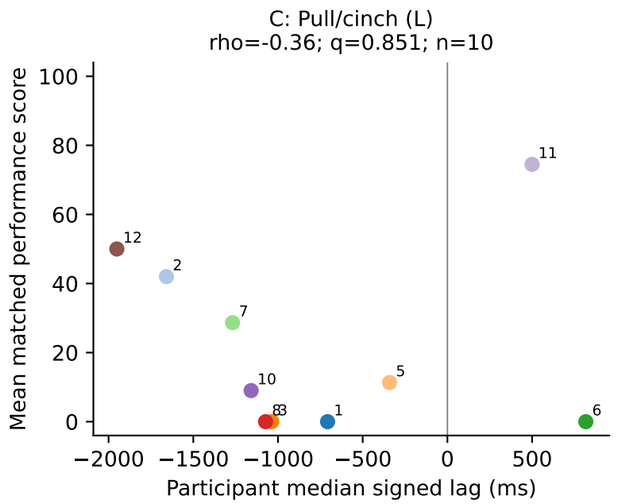






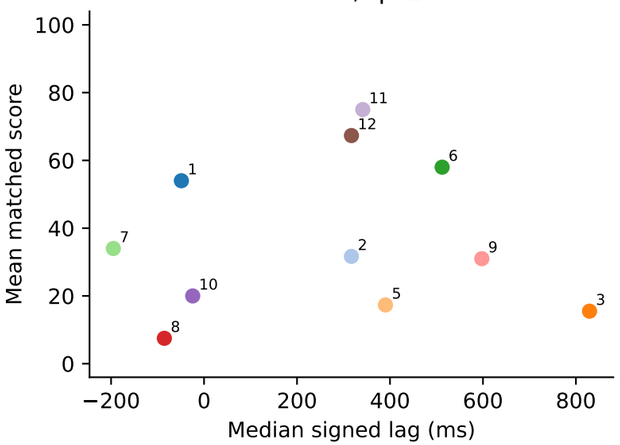




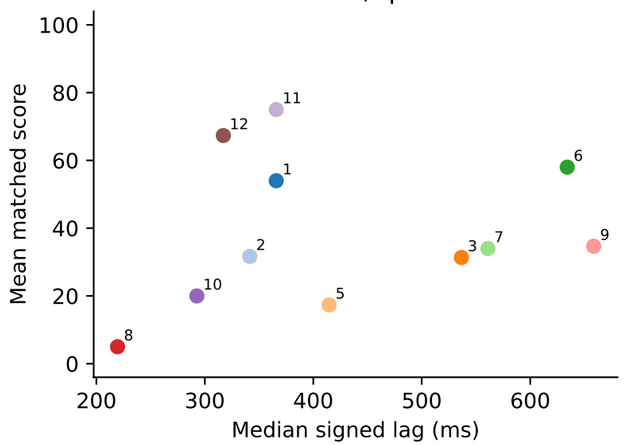


Whole-task peak features vs performance: participant summaries

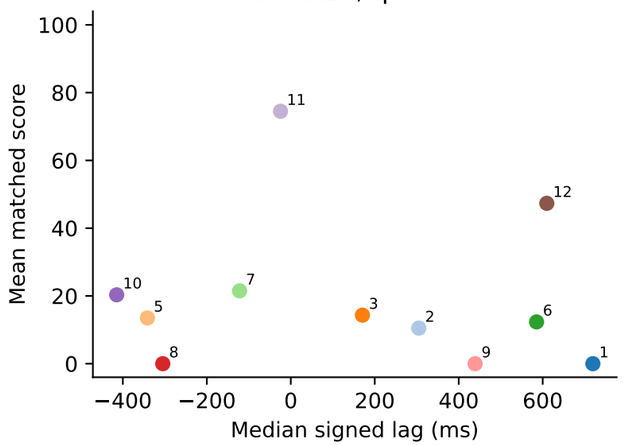
A: Whole task (L)
rho=-0.05; q=1.000



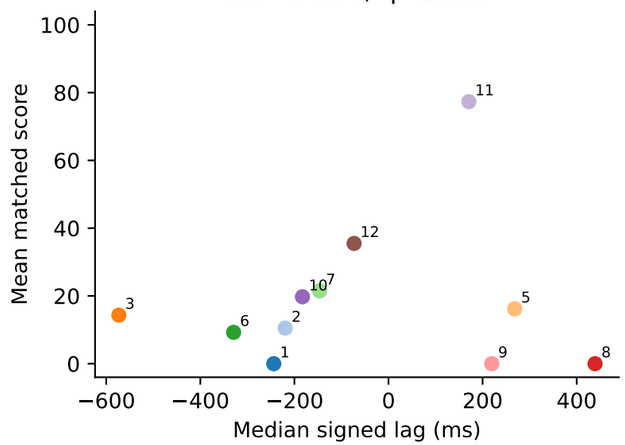
A: Whole task (R)
rho=+0.30; q=0.851



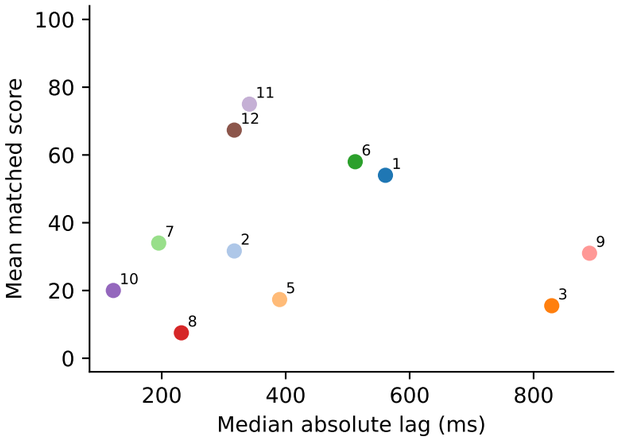
C: Whole task (L)
rho=-0.24; q=0.857



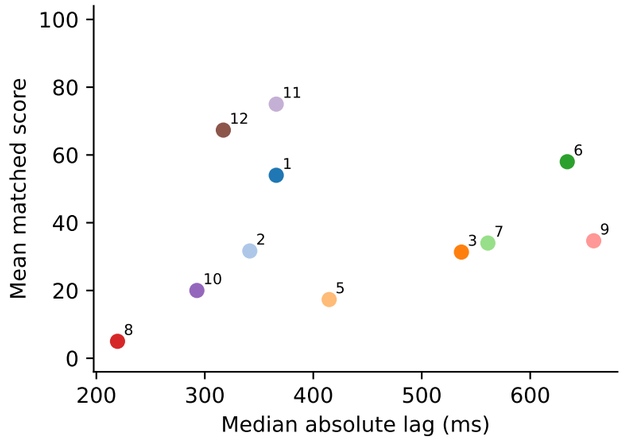
C: Whole task (R)
rho=+0.06; q=1.000



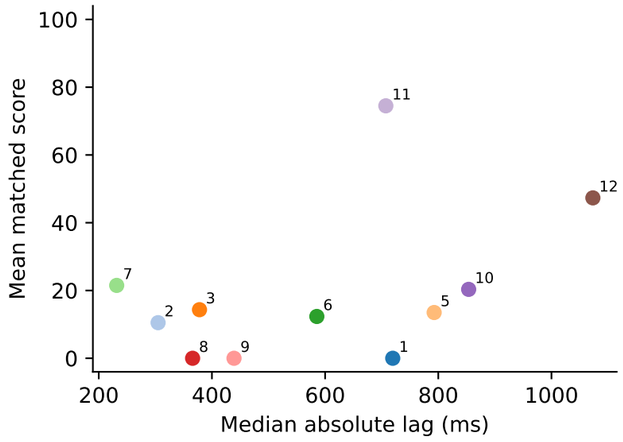
A: Whole task (L)
rho=+0.03; q=1.000



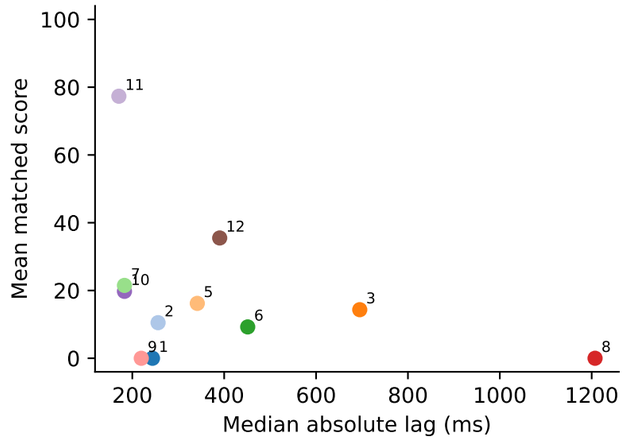
A: Whole task (R)
rho=+0.30; q=0.851



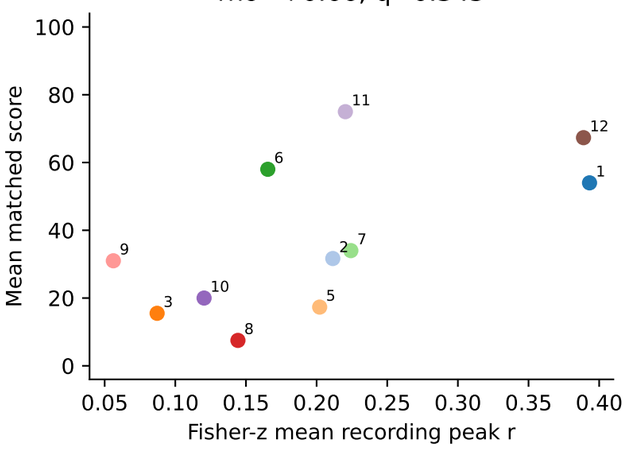
C: Whole task (L)
rho=+0.29; q=0.851



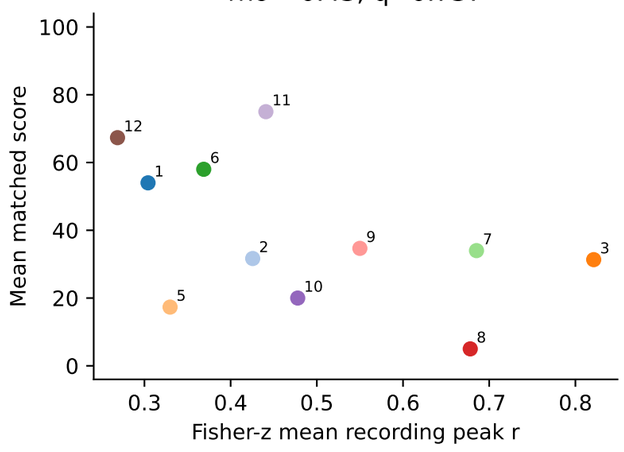
C: Whole task (R)
rho=-0.44; q=0.737



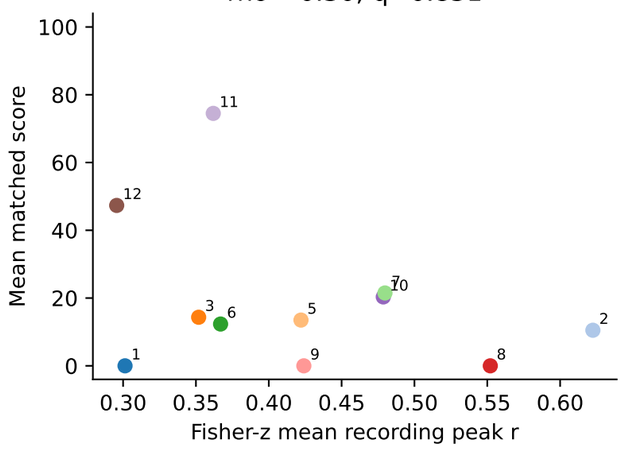
A: Whole task (L)
rho=+0.66; q=0.345



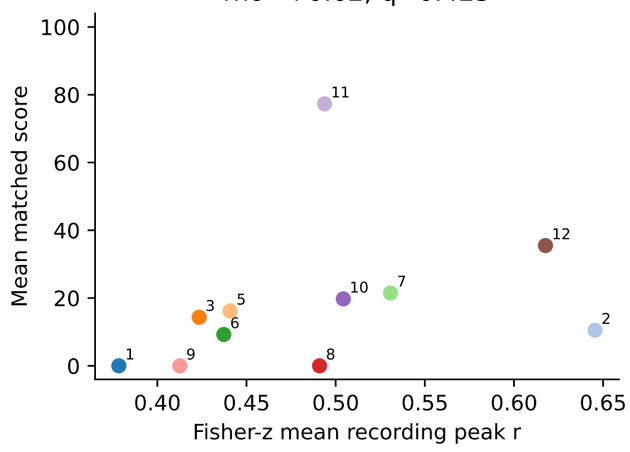
A: Whole task (R)
rho=-0.43; q=0.737

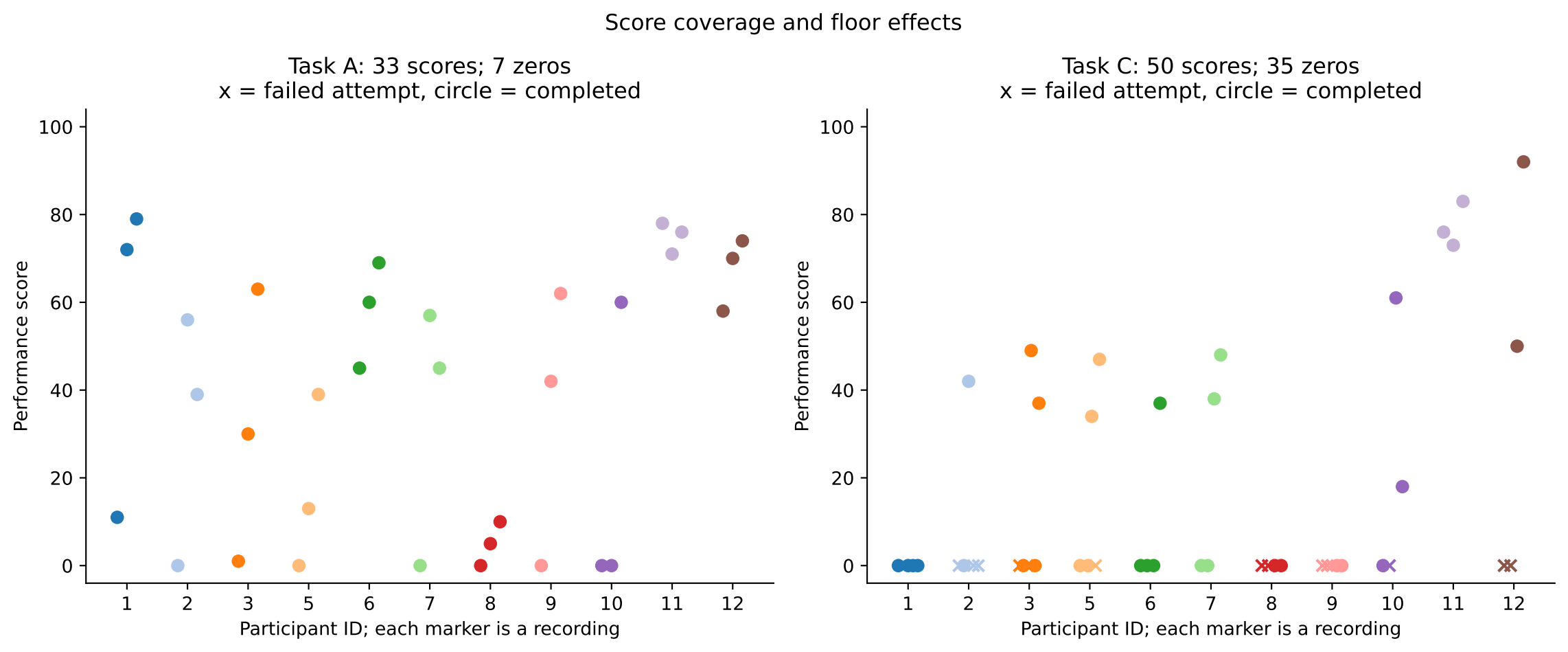


C: Whole task (L)
rho=-0.30; q=0.851

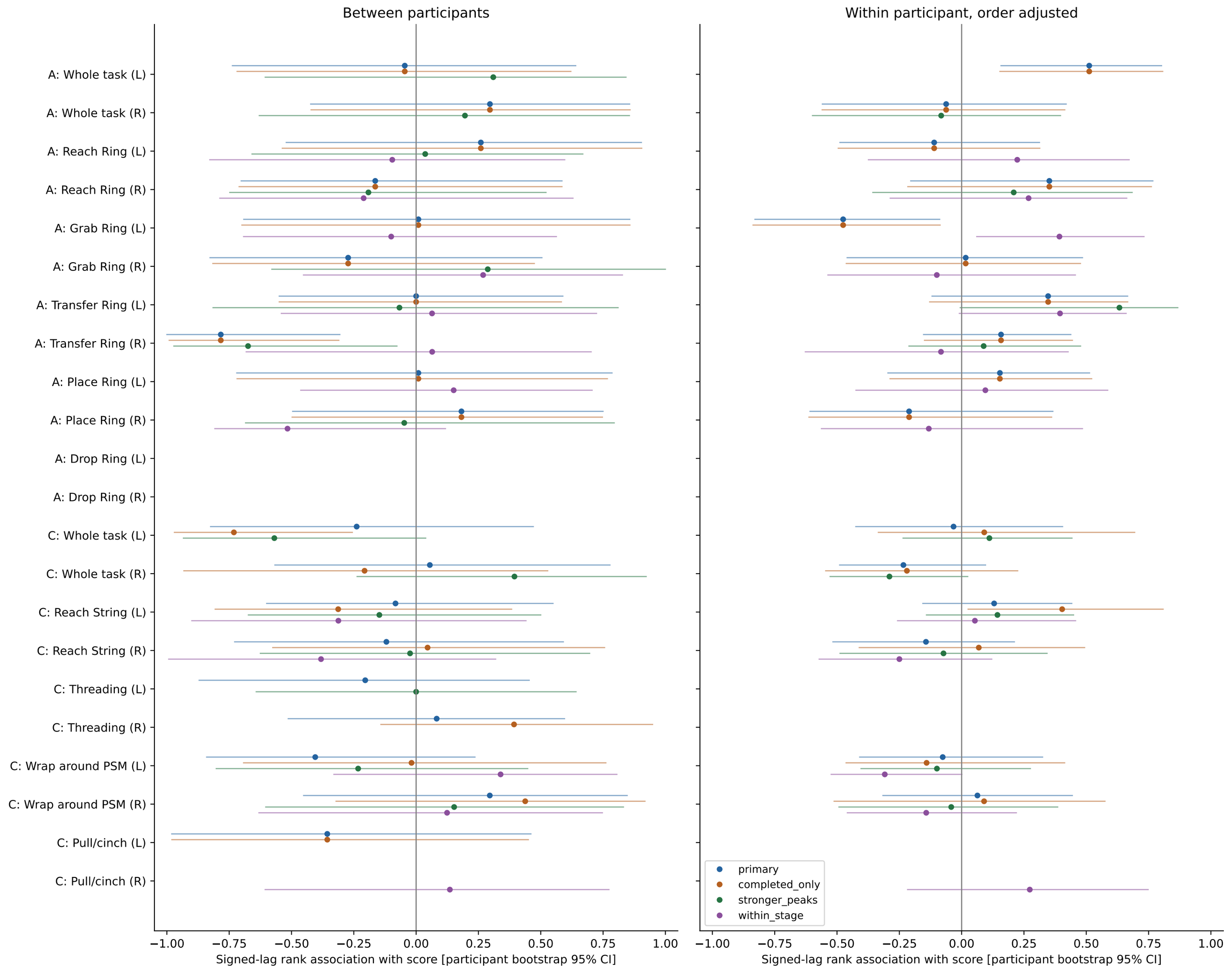


C: Whole task (R)
rho=+0.62; q=0.423




Score coverage and floor effects
Task A: 33 scores; 7 zeros
x = failed attempt, circle = completed
Task C: 50 scores; 35 zeros
x = failed attempt, circle = completed
Performance score
Participant ID; each marker is a recording
0
20
40
60
80
100
1
2
3
5
6
7
8
9
10
11
12

Timing association sensitivity
Missing estimates indicate an excluded group or fewer than six informative participants.



Secondary velocity result: correlation strength during Task A grab ring
These associations describe strength, not peak lead/lag; order adjustment weakens the evidence.

